

Section I**Attempt Questions 1-10****All questions are equal value.****Use the multiple choice answer sheet for Questions 1-10**

1 Simplify $\frac{x^3 - 1}{x^2 - 1} \times \frac{x^2 - 4x - 5}{4x^2 + 4x + 4}$

(A) $\frac{(x-5)}{4}$

(B) $\frac{(x-1)}{4}$

(C) $\frac{(x+1)}{4}$

(D) $\frac{(x^2 + x + 1)}{4}$

2 What is the solution to the equation $|x - 2| = 2x - 1$?

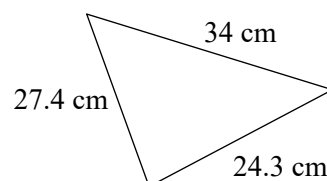
(A) $x = -3$

(B) $x = -1$

(C) $x = 1$

(D) $x = 3$

3 The smallest angle in the triangle below is θ .



What is the value of θ to the nearest degree?

(A) 30°

(B) 45°

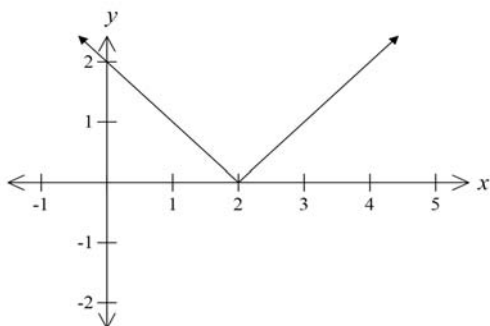
(C) 53°

(D) 82°

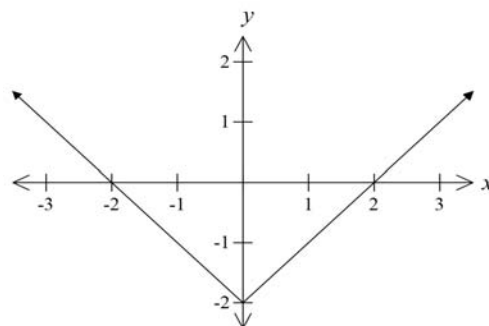
The smallest angle in the triangle below is θ .

4 Which graph best represents $y = |x| - 2$?

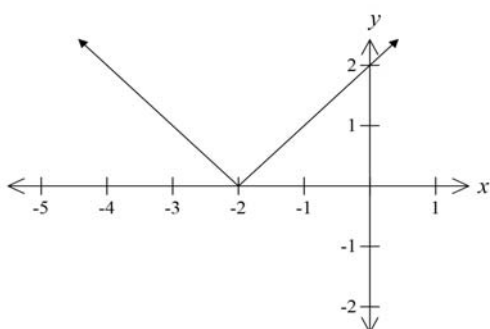
(A)



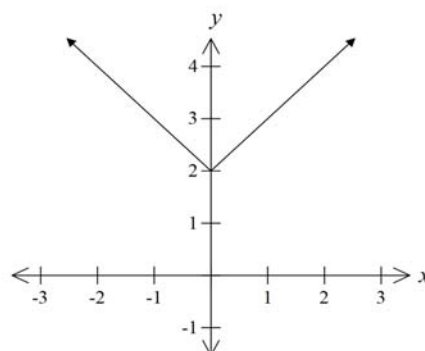
(B)



(C)



(D)



5 Which of these is the limiting sum of the geometric series $\frac{2}{5} - \frac{2}{15} + \frac{2}{45} - \frac{2}{135} + \dots$

(A) $\frac{3}{5}$

(B) $\frac{8}{27}$

(C) 0

(D) $\frac{3}{10}$

6 If $3\cos\theta + 2 = 0$ and $\tan\theta > 0$, what is the exact value of $\sin(\theta + 180^\circ)$?

(A) $-\frac{\sqrt{5}}{3}$

(B) $-\frac{\sqrt{5}}{2}$

(C) $\frac{\sqrt{5}}{2}$

(D) $\frac{\sqrt{5}}{3}$

- 7 What is the centre and radius of the circle with the equation $x^2 + y^2 + 6x - 8y - 11 = 0$
- (A) Centre $(-3, -4)$ and radius 36
(B) Centre $(-3, 4)$ and radius 36
(C) Centre $(-3, -4)$ and radius 6
(D) Centre $(-3, 4)$ and radius 6
- 8 What is the value of k if the sum of the roots of $x^2 - (k-1)x + 2k = 0$ is equal to the product of the roots?
- (A) -3
(B) -2
(C) -1
(D) 1
- 9 Which of the following is the correct simplified expression for differentiating $f(x) = \frac{1}{x}$ from first principles?
- (A) $f'(x) = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)}$
(B) $f'(x) = \lim_{h \rightarrow 0} \frac{x+h-x}{h}$
(C) $f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x} - \frac{1}{x+h}}{h}$
(D) $f'(x) = \lim_{h \rightarrow 0} \frac{h}{x+h-x}$
- 10 What is the equation of the normal to the curve $f(x) = x^2 - 4x$ at $(1, -3)$?
- (A) $x + 2y - 7 = 0$
(B) $x - 2y - 7 = 0$
(C) $2x - y - 5 = 0$
(D) $2x + y + 5 = 0$

Section II**Attempt Questions 11-14****Each question is worth 15 marks.****Answer each question in a new writing booklet. Extra booklets are available.****All necessary working should be shown in every question**

Question 11 (15 Marks) Use a NEW writing booklet.

- (a) Solve the following equation $x^2 + 3x = \frac{8}{x^2 + 3x} + 2$
by using the substitution $A = x^2 + 3x$. **3**
- (b) Find A, B and C such that: **3**
$$4x^2 - x + 1 \equiv Ax(x+1) + B(x+1) + C.$$
- (c) Solve $x^6 + 26x^3 - 27 = 0$ **2**
- (d) Solve for x $\frac{2}{x-1} \geq x$. **3**
- (e) What are the coordinates of the point that divides the interval joining the points $A(1,1)$ and $B(5,3)$ externally in the ratio 2:3? **2**
- (f) Find the equation of the straight line that passes through the point of intersection of the lines $x - 2y = 5$ and $3x - y + 1 = 0$ and the point $(2,1)$. **2**

Question 12 (15 Marks) Use a NEW writing booklet.

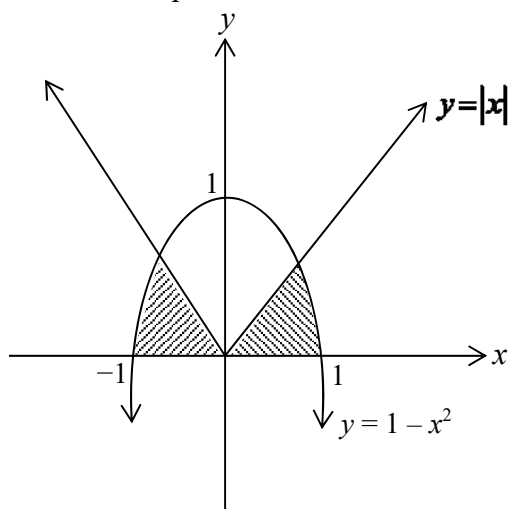
(a) State the domain of:

$$(i) \quad y = x + \frac{1}{x-2} \quad 1$$

$$(ii) \quad y = \sqrt{2x^2 - x - 6} \quad 2$$

(b) Find the horizontal asymptote of the function $y = \frac{2x^2 - 4x + 3}{x^2 - 5}$ 2

(c) Find the inequalities that describe the shaded regions in the following graph. 3



**Diagram
not to
scale**

(d) Shade the common region defined by:

$$x^2 + y^2 < 25 \quad \text{and} \quad 3x - y \geq 2 \quad 3$$

(e) What values of m is $-4x^2 + 3x + m$ a negative definite? 2

(f) For what values of c is the line $y = x + c$ tangent to the curve $y = 2x^2 - 7x + 4$ 2

Question 13 (15 Marks) Use a NEW writing booklet.

(a) Evaluate $\sum_{n=0}^{20} (-2)^n$ **3**

(b) Find the sum of all positive integers less than 20 000 which are divisible by 11. **3**

(c) Prove by mathematical Induction

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$
4

(d) For the parabola $x^2 = -16y + 32$.

(i) Give the vertex and focus of the parabola. **3**

(ii) Find the equation of the tangent to the parabola at the point (8, -2). **2**

Question 14 (15 Marks) Use a NEW writing booklet.

(a) Solve $2\cos 2\theta = \sqrt{3}$ for $0^\circ \leq \theta \leq 360^\circ$. **3**

(b) Prove the identity $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \sec x \cdot \csc x$ **2**

(c) (i) Show that $10\sin^2 \beta + \cos \beta - 7 = (3 - 5\cos \beta)(2\cos \beta + 1)$. **2**

(ii) Hence solve $10\sin^2 \beta + \cos \beta - 7 = 0$ for $0^\circ < \beta < 360^\circ$ to the nearest degree. **3**

(d) Differentiate $\sqrt[3]{2-3x^2}$, give your answer without negative or fractional indices. **2**

(e) Find the value of k if $f'(-3) = 1$ where **3**

$$f(x) = \frac{x^2 + k}{x^2 - k}.$$

END of PAPER

Yr 11 Ext 1 Prelim 2012

1	A
2	C
3	B
4	B
5	D
6	D
7	D
8	C
9	A
10	B

Ext1 Preliminary Examination 2012

Question 11

$$(a) \frac{x^2 + 3x}{x^2 + 3x} = \frac{8}{x^2 + 3x} + 2$$

$$\text{let } M = x^2 + 3x$$

$$M = \frac{8}{M} + 2$$

$$M^2 = 8 + 2M$$

$$M^2 - 2M - 8 = 0$$

$$(M - 4)(M + 2) = 0$$

$$\therefore M = 4 \text{ or } -2 \quad \checkmark$$

$$\therefore x^2 + 3x = 4 \text{ or } x^2 + 3x = -2$$

$$x^2 + 3x - 4 = 0 \quad x^2 + 3x + 2 = 0$$

$$(x + 4)(x - 1) = 0 \quad (x + 2)(x + 1) = 0$$

$$\therefore x = -4 \text{ or } 1 \quad \checkmark \quad \therefore x = -2 \text{ or } -1 \quad \checkmark$$

3 marks

$$(b) 4x^2 - x + 1 \equiv Ax(x+1) + B(x+1) + C$$

Method 1

$$\equiv Ax^2 + Ax + Bx + B + C$$

$$\equiv Ax^2 + (A+B)x + (B+C)$$

$$\therefore 4x^2 \equiv Ax^2$$

$$4 = A \quad \checkmark$$

$$-x = (A+B)x$$

$$-1 = (4+B)$$

$$\therefore -5 = B \quad \checkmark$$

$$1 = B + C$$

$$1 = -5 + C$$

$$6 = C \quad \checkmark$$

Method 2

$$A = 4 \quad \checkmark$$

$$\text{let } x = -1 \quad 6 \equiv 0 + 0 + C$$

$$6 = C \quad \checkmark$$

$$\text{let } x = 0$$

$$1 \equiv 0 + B + C$$

$$1 \equiv B + 6$$

$$\therefore -5 = B \quad \checkmark$$

$$\therefore A = 4, B = -5, C = 6$$

3 marks

$$(c) x^6 + 26x^3 - 27 = 0$$

$$\text{let } u = x^3$$

$$u^2 + 26u - 27 = 0$$

$$(u+27)(u-1) = 0$$

$$\therefore u = -27 \text{ or } u = 1 \quad \checkmark$$

$$\therefore x^3 = -27 \text{ or } x^3 = 1$$

$$\therefore x = -3 \text{ or } 1 \quad \checkmark \quad (2 \text{ marks})$$

$$(d) \frac{2}{x-1} \geq x \quad x \neq 1$$

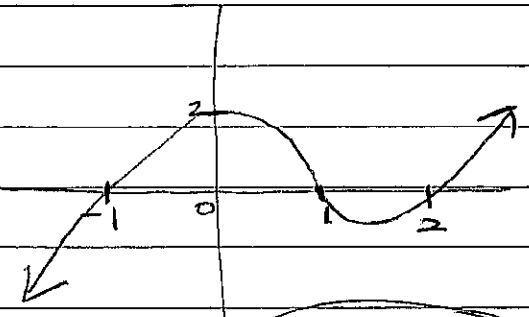
$$2(x-1) \geq x(x-1)^2 \quad \checkmark$$

$$0 \geq x(x-1)^2 - 2(x-1)$$

$$0 \geq (x-1)(x(x-1)-2)$$

$$0 \geq (x-1)(x^2-x-2)$$

$$0 \geq (x-1)(x-2)(x+1) \quad \checkmark$$



3 marks

$$\therefore x \leq -1 \text{ or } 1 \leq x \leq 2 \quad x \neq 1$$

$$\therefore x \leq -1 \text{ or } 1 < x \leq 2 \quad \checkmark$$

$$(e) A(1,1) \quad B(5,3)$$

$$k: 2$$

$$2: -3 \quad \text{External}$$

$$P(x,y) = \left(\frac{kx_2 + 1x_1}{k+1}, \frac{ky_2 + 1y_1}{k+1} \right)$$

$$= \left(\frac{2 \times 5 + 1 \times 1}{-1}, \frac{2 \times 3 + 1 \times 1}{-1} \right) \quad \checkmark$$

$$= \left(\frac{10+1}{-1}, \frac{6+1}{-1} \right)$$

$$= (-7, -3) \quad \checkmark$$

2 marks

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Question 11 Continued.

$$(f) \quad x - 2y = 5$$

$$3x - y + 1 = 0$$

$$(x - 2y - 5) + k(3x - y + 1) = 0 \quad \text{sub in } (2, 1)$$

$$(2 - 2 - 5) + k(6 - 1 + 1) = 0$$

$$-5 + k \times 6 = 0$$

$$6k = 5$$

$$k = \frac{5}{6} \checkmark$$

$$\therefore (x - 2y - 5) + \frac{5}{6}(3x - y + 1) = 0$$

$$6(x - 2y - 5) + 5(3x - y + 1) = 0$$

$$6x - 12y - 30 + 15x - 5y + 5 = 0$$

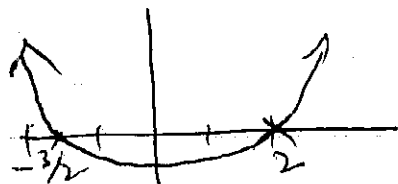
$$\therefore 21x - 17y - 25 = 0 \checkmark \quad (2 \text{ marks}).$$

Method 2 - if the equations were solve simultaneously
point of intersection $(-\frac{7}{5}, -\frac{16}{5}) \checkmark$

$$\text{equation of line } 21x - 17y - 25 = 0 \checkmark$$

Question 12 (15) Ext 1 Y11 Prelim

(a) (i) $x \in \mathbb{R}, x \neq 2$ ✓
 (ii) $2x^2 - 7x - 6 \geq 0$ ✓
 $(2x+3)(x-2) \geq 0$



$x \leq -\frac{3}{2}, x \geq 2$ ✓

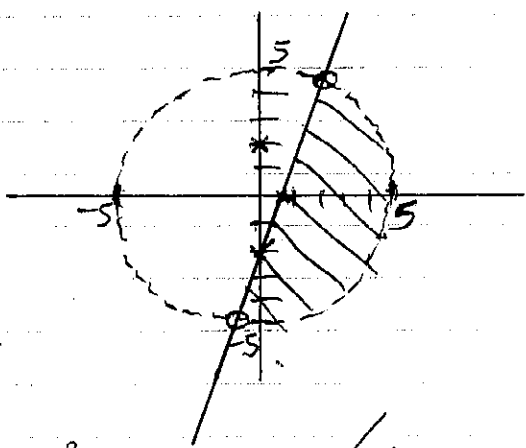
(b) $y = \frac{2 - \frac{4}{x} + \frac{3}{x^2}}{1 - 5/x}$ ✓

As $x \rightarrow \infty$ $y \rightarrow \frac{2}{1}$.

∴ horizontal asymptote is $y = 2$ ✓

(c) $y \leq 1 - x^2, y \leq |x|, y \geq 0$ ✓

(d)



1 broken circle
 1 unbroken circle
 1 right of correct line.

(e) $b^2 - 4ac < 0$ ✓ (and $a < 0$, which it is as $a = -4$)

$9 - 4x - 4xm < 0$

$9 + 16m < 0$

$16m < -9$

$m < -\frac{9}{16}$ ✓

(f) solve $x+c = 2x^2 - 7x + 4$ ✓
 $2x^2 - 8x + (4-c) = 0$

largest when $\Delta = 0$.

$64 - 4 \times 2 \times (4-c) = 0$

$64 - 8(4-c) = 0$

$64 - 32 + 8c = 0$ ✓

$c = -4$

Yr 11 Ext 1 Prelim 2012

Section I

1. A 2. C 3. B 4. B 5. D 6. D 7. D 8. C 9. A 10. B

Section II

Question 13

a) $\sum_{n=0}^{20} (-2)^n = 1 + -2 + 4 + -8 + \dots$

$a = 1 \quad r = -2 \quad n = 21$ ✓

$S_{21} = \frac{1(-2^{21}-1)}{-2-1}$ ✓

$= 699051$ ✓

b) $11 + 22 + 33 + \dots$

$T_n = a + (n-1)d$

$20000 > 11 + (n-1)11$

$20000 > 11 + 11n - 11$

$n < 1818.18$ ✓

$n = 1818$ ✓

$S_{1818} = \frac{1818}{2} (22 + 1817 \times 11)$

$= 18188181$ ✓

c) Step 1
Prove true for $n=1$

$LHS = \frac{1}{(2(1)-1)(2(1)+1)} = \frac{1}{1 \times 3} = \frac{1}{3}$

$RHS = \frac{1}{2(1)+1} = \frac{1}{3}$

$LHS = RHS$

$\therefore$ true for $n=1$

Step 2 Assume true for $n=k$

$\sum_{n=1}^k \frac{1}{(2n-1)(2n+1)} = \frac{k}{2k+1}$ ✓

notation
must be
correct

Step 3 Prove true for $n = k+1$

$$\text{Prove } \sum_{r=1}^{k+1} \frac{1}{(2r-1)(2r+1)} = \frac{k+1}{2(k+1)+1}$$

$$\text{LHS} = \sum_{r=1}^{k+1} \frac{1}{(2r-1)(2r+1)}$$

$$= \sum_{r=1}^k \frac{1}{(2r-1)(2r+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \quad \checkmark$$

$$= \frac{k}{2k+1} + \frac{1}{(2k+1)(2(k+1)+1)} \quad \checkmark \quad \text{From Step 2}$$

$$= \frac{k(2(k+1)+1) + 1}{(2k+1)(2(k+1)+1)}$$

$$= \frac{2k^2 + 3k + 1}{(2k+1)(2(k+1)+1)}$$

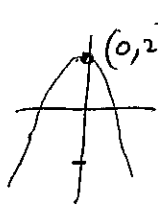
$$= \frac{(k+1)(2k+1)}{(2k+1)(2(k+1)+1)} \quad \checkmark$$

$$= \frac{k+1}{2(k+1)+1}$$

$\therefore$ true for $n = k+1$

Step 4

d) i) $x^2 = -16y + 32$



$x^2 = -16(y-2) \quad \checkmark$
 $a = 4 \quad \text{vertex } (0, 2) \quad \checkmark$
 $\text{focus } (0, -2) \quad \checkmark$

ii) $-16y = x^2 - 32$

$$y = -\frac{1}{16}x^2 + 2$$

$$y' = -\frac{1}{8}x$$

$$x = 8 \quad m = -1 \quad \checkmark$$

$$y - 2 = -1(x - 8)$$

$$y + 2 = -x + 8$$

$$y = -x + 6 \quad \checkmark$$

Q14(a) $\cos 2\theta = \frac{\sqrt{3}}{2}$

$$2\theta = 30^\circ, 330^\circ, 390^\circ, 690^\circ$$

$$\theta = 15^\circ, 165^\circ, 195^\circ, 345^\circ$$

(b) LHS = $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}$
 $= \frac{\sin^2 x + \cos^2 x}{\cos x \sin x}$
 $= \frac{1}{\cos x \sin x}$
 $= \sec x \operatorname{cosec} x$
 $= \text{RHS}$

(c)(i) $10(1 - \cos^2 \beta) + \cos \beta - 7$
 $= 10(1 - x^2) + x - 7, x = \cos \beta$
 $= -(10x^2 - x - 3)$
 $= -(5x - 3)(2x + 1)$
 $= (3 - 5x)(2x + 1), x = \cos \beta$
(ii) $(3 - 5x)(2x + 1) = 0, x = \cos \beta$
 $x = \frac{3}{5} \text{ or } -\frac{1}{2}$
 $\cos \beta = \frac{3}{5} \text{ or } -\frac{1}{2}$
 $\beta = 53^\circ, 307^\circ, 120^\circ, 240^\circ$
 $\nwarrow$ nearest degree

(d) $\frac{d}{dx} (2 - 3x^2)^{\frac{1}{3}} = \frac{1}{3} (2 - 3x^2)^{-\frac{2}{3}} \cdot -6x$
 $= \frac{-2x}{\sqrt[3]{(2 - 3x^2)^2}}$

(e) $f(x) = \frac{x^2 + k}{x^2 - k}$
 $f'(x) = \frac{(x^2 - k) \cdot 2x - (x^2 + k) \cdot 2x}{(x^2 - k)^2}$
 $= \frac{-4kx}{(x^2 - k)^2}$

$f(-3) = \frac{12k}{(9 - k)^2} = 1$

$$12k = (9 - k)^2 = 81 - 18k + k^2$$

$$k^2 - 30k + 81 = 0$$

$$(k - 3)(k - 27) = 0, k = 3 \text{ or } 27$$

- ① 1 correct
 ② 2 "
 ③ All "

①

①

① $1 - \cos^2 \beta$

②

①

① 2 solutions

① All "

①

②

①

①

①